Adversarial Forecasting with LSTM vs GAN-LSTM Models: A Comparative Analysis

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***Abstract*—Long Short-Term Memory (LSTM) networks are widely used for wind-speed forecasting, yet their autoregressive multi-step predictions often degrade at longer horizons due to error accumulation and loss of temporal structure. This paper introduces a lightweight adversarial refinement strategy that augments a pretrained LSTM forecaster with a discriminator trained to evaluate the realism of single-step wind-speed predictions conditioned on recent history. Unlike full GAN-based forecasting frameworks, the proposed approach preserves the original forecaster architecture and applies adversarial feedback only at the next-step level. The method is evaluated on daily multivariate weather data using multi-horizon autoregressive rollouts. Beyond mean absolute error (MAE), structural metrics including autocorrelation function (ACF) cosine similarity and forecast sharpness are used to assess temporal coherence and dynamical stability. Results show that while short-term accuracy remains comparable, the adversarially refined LSTM substantially improves temporal structure and stability at medium and long horizons, mitigating structural collapse observed in the baseline model and producing more dynamically realistic multi-day forecasts.**

***Index Terms*—Time-series forecasting, LSTM, generative adversarial network, rare event prediction, temporal coherence, multivariate time series**

I. Introduction

Accurate wind-speed forecasting is crucial for modern energy systems. It aids in applications like wind farm scheduling, grid stability, and integrating renewable energy [1]. Recently, deep learning models, particularly Long Short-Term Memory (LSTM) networks [2], have gained popularity for time-series forecasting. They effectively model complex, nonlinear time dependencies. However, while LSTMs usually perform well in the short term, their accuracy tends to decline over longer periods. Small prediction errors can accumulate, which can result in a loss of time structure, excessive smoothing, or unstable dynamics [3].

Recent studies have explored the use of generative adversarial networks (GANs) [4] for time-series modeling. These models can capture complex data distributions and maintain realistic variability. Nevertheless, complete GAN-based forecasting systems can be hard to train, require substantial computational power, and are sensitive to hyperparameter settings [5]. These challenges limit their practical applications in real-world forecasting. Many studies primarily focus on pointwise accuracy metrics, providing minimal insight into whether long-range time structure is preserved.

In this work, we present a lightweight adversarial refinement strategy. This approach improves a standard LSTM forecaster by adding a discriminator to assess the realism of one-step predictions. Instead of replacing the forecasting model or generating entire sequences with adversarial methods, our strategy reuses a pretrained LSTM as the generator. It provides adversarial feedback only at the next-step prediction level. This maintains the simplicity and efficiency of standard LSTM forecasting while using adversarial learning to enhance the realism and stability of autoregressive predictions.

To evaluate the effectiveness of this refinement, we conduct a thorough multi-horizon analysis using daily multivariate weather data. In addition to traditional error metrics like mean absolute error (MAE), we use structural evaluation measures such as autocorrelation function (ACF) cosine similarity and forecast sharpness [6]. These measures help assess temporal coherence and dynamic behavior. Our experimental results show that the adversarially refined model consistently reduces error accumulation, preserves time dependency structure at medium ranges, and produces smoother yet dynamically realistic long-horizon forecasts. Specifically, the refined model shows significant improvement in ACF similarity at the five-day horizon, addressing the structural collapse seen in the baseline LSTM.

The main contributions of this paper are threefold: (i) introducing an efficient adversarial refinement mechanism for LSTM-based forecasting, (ii) providing a detailed empirical analysis of how adversarial feedback influences long-horizon autoregressive behavior, and (iii) creating an evaluation framework that considers both pointwise accuracy and temporal structure together. These contributions offer fresh insights into how selective adversarial learning can enhance the realism and robustness of time-series forecasts without the complexity of complete generative models.

II. Related Work

Time-series forecasting has been widely studied in areas like energy systems, meteorology, and finance. Traditional statistical methods, including autoregressive integrated moving average (ARIMA) models [7] and state-space formulations [8], have long been used for wind and weather prediction [9], [10]. While these methods offer interpretability and theoretical guarantees, they often fail to capture nonlinear dependencies and

complex temporal patterns found in real-world atmospheric data.

Deep learning models, especially recurrent neural networks (RNNs) and LSTM networks [2], have shown strong performance in short-term forecasting tasks because they can model nonlinear time relationships [3], [11]. Many studies have applied LSTMs to wind-speed and wind-power forecasting, reporting better accuracy than traditional methods [1], [12]–[14]. However, it is well-known that LSTM-based forecasters face issues with error accumulation and loss of time coherence in autoregressive multi-step settings [15], [16], leading to overly smooth or unstable long-horizon predictions [3].

Generative adversarial networks (GANs) [4] have been suggested as a way to tackle some of these challenges by promoting forecasts that better align with the distributional properties of real data. Variants like TimeGAN [5] and sequence-to-sequence GAN frameworks [17], [18] have been explored for time-series generation and forecasting. While these approaches often enhance realism and variability, they usually involve complex training processes, multiple discriminators, or sequence-level adversarial losses [19], making them computationally expensive and sensitive to hyperparameter tuning. Additionally, many GAN-based forecasting studies focus on sample realism or distribution matching instead of directly evaluating long-horizon time structure in autoregressive forecasts.

Recent work has pointed out the significance of structural evaluation metrics beyond pointwise error, which include measures of time dependency, autocorrelation, and volatility [6], [20], [21]. These metrics provide insight into whether a model maintains the underlying dynamics of the time series, especially over long periods. However, relatively few studies systematically combine adversarial learning with structural evaluation to examine how temporal coherence degrades as the forecast horizon extends.

This work stands out by presenting a lightweight adversarial refinement strategy that enhances a standard LSTM forecaster without changing its structure or training a complete generative model. Adversarial feedback is applied only to single-step predictions, allowing the refinement to influence long-range behavior through improved local transition dynamics. Unlike earlier GAN-based methods, this approach prioritizes computational efficiency, stability, and interpretative evaluation. It combines traditional accuracy metrics with structural measures like ACF similarity and forecast sharpness for a more thorough assessment of long-horizon forecasting performance.

III. Methodology

This section describes the complete forecasting pipeline, including data preprocessing, sliding-window sequence construction, baseline LSTM training, adversarial refinement, and multi-horizon evaluation. Figure 1 provides an overview of the full workflow

Dataset

The dataset consists of daily multivariate meteorological observations obtained from a publicly available Kaggle weather dataset. Each record contains a DATE field along with several atmospheric and indicator variables. The available features include:

- **WIND**: average daily wind speed (knots), used as the forecasting target,
- **RAIN**: daily precipitation amount (mm),
- **T.MAX**: maximum temperature (°C),
- **T.MIN**: minimum temperature (°C),
- **T.MIN.G**: 09:00 UTC grass minimum temperature (°C),
- **IND**, **IND.1**, **IND.2**: auxiliary indicator variables supplied with the dataset.

All non-date columns are retained as candidate predictors.The dataset spans [X] days from [start date] to [end date], obtained from [specific Kaggle dataset name/URL], representing meteorological observations from [location]. After removing missing values, [Y] complete daily records remained.

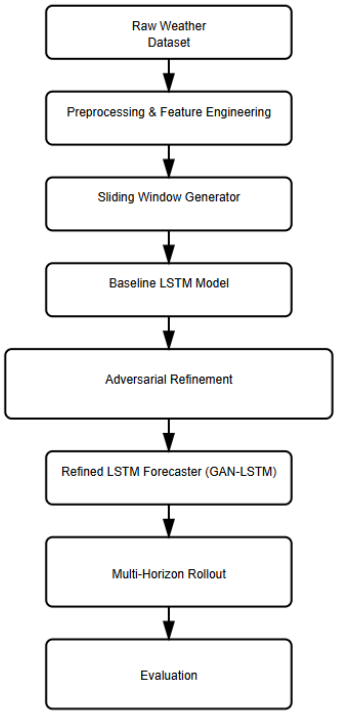

Fig. 1. Overall forecasting pipeline from raw weather data to adversarially refined GAN-LSTM forecasts and multi-horizon evaluation.

Parsing and cleaning

DATE is parsed as a datetime index, the table is sorted chronologically, and rows with missing values in any feature are removed. No interpolation or duplicate handling is applied beyond this NaN filtering.

Cyclic encodings

To capture seasonality, we engineer daily and yearly cyclic features: day of week sine/cosine and day of year sine/cosine, computed directly from DATE. The final feature set is the original non-date columns plus these four encodings.

Scaling

All features are standardized to zero mean and unit variance using StandardScaler fit on the cleaned dataset; inverse transforms are applied at evaluation to report metrics in original units.

Temporal split

Sliding windows are built on the scaled series, then partitioned chronologically into training (70%), validation (15%), and testing (15%) without shuffling, preserving temporal order to avoid leakage.

Supervised Sequence Construction

Following preprocessing and feature normalization, the multivariate daily time series is converted into supervised learning samples suitable for recurrent neural networks. A sliding-window approach is used: for a chosen history length N, each input sequence consists of N consecutive days of standardized features, and the corresponding target is the feature vector of the following day.

Formally, let $x_t \in \mathbb{R}^F$ denote the standardized feature vector on day t. Each training sample is constructed as

$$X^{(i)} = [\, x_i,\ x_{i+1},\ \ldots,\ x_{i+N-1} \,] \in \mathbb{R}^{N \times F},$$

with the supervised target

$$y^{(i)} = x_{i+N}.$$

This transforms the original time series of length M into $M - N$ supervised pairs

$$\big(X^{(i)},\ y^{(i)}\big).$$

These window–target pairs form the datasets X_{train}, X_{val}, X_{test} and their corresponding next-day targets. Only chronological ordering is used when segmenting the data; no shuffling is applied. This ensures that the model is trained strictly on past information and evaluated on unseen future periods, matching real-world forecasting conditions.

Baseline LSTM Forecasting Model

The first forecasting model examined in this study is a standard Long Short-Term Memory (LSTM) network trained to predict the next-day feature vector from a fixed-length history window. This architecture serves as the non-adversarial baseline against which the refined model is evaluated.

The model receives an input sequence $X \in \mathbb{R}^{N \times F}$, where N denotes the number of past days and F the number of standardized features. A single LSTM layer processes the temporal dimension of the input and produces a fixed-size hidden representation summarizing the history window. To reduce overfitting and improve generalization, a dropout layer is applied to the LSTM output during training. The final prediction is generated by a fully connected layer that maps the hidden state to an F-dimensional output vector corresponding to the next-day feature estimate.

Training is performed using the Adam optimizer with a weighted mean squared error (MSE) loss. The weighting emphasizes the wind-speed component of the prediction to account for its importance in downstream evaluation, while still allowing the network to model interactions among all features. Early stopping based on validation loss is applied to prevent overtraining and to ensure stable convergence. Once trained, the baseline model is used both for next-day prediction and as the generator component in the adversarial refinement stage described later.

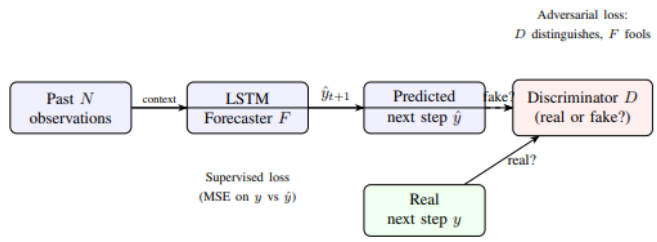

Fig. 2. GAN-LSTM training framework showing the interaction between the forecaster F and discriminator D during adversarial refinement. The discriminator evaluates real and predicted next-day wind values, while the forecaster is refined through a combination of supervised and adversarial losses.

Adversarial Refinement Framework

To address the well-known tendency of autoregressive LSTM forecasters to lose temporal structure during long-horizon rollouts, a lightweight adversarial refinement stage is introduced after training the baseline model. Rather than constructing a full generative adversarial network, the approach reuses the pretrained LSTM forecaster as the generator and trains a small discriminator network to evaluate the realism of its next-day wind-speed predictions given the recent history window. The objective is to subtly adjust the forecaster so that its single-step predictions better match the statistical properties of real atmospheric dynamics, which in turn improves stability and realism over multi-step rollouts.

The discriminator operates on two inputs: (i) the N-day history window $X \in \mathbb{R}^{N \times F}$, and (ii) a candidate next-day wind value (real or predicted). The history window is summarized using a bidirectional LSTM followed by dense layers, producing a contextual embedding that captures the recent temporal pattern. This embedding is concatenated with the candidate wind value, and the resulting vector is passed through additional dense layers to output a probability representing how likely the proposed next-day value is to be real.

During adversarial refinement, the discriminator is trained to distinguish ground-truth next-day wind values from those produced by the baseline forecaster. Conversely, the forecaster is updated to minimize a combined loss consisting of (i) a supervised mean squared error (MSE) term to preserve predictive accuracy, and (ii) an adversarial term encouraging the discriminator to classify the predicted next-day value as real. A weighting parameter λ controls the influence of the adversarial signal. Training proceeds in alternating steps: the discriminator is first updated using real and generated samples, followed by an update to the forecaster using the combined

objective. A brief supervised "warm-up" stage precedes adversarial refinement to stabilize next-day predictions before adversarial feedback is introduced.

This refinement strategy is deliberately lightweight: only the next-step wind component is adversarially evaluated, the forecaster architecture remains unchanged, and training is performed for a small number of epochs. Despite its simplicity, this procedure improves multi-day behavior, reducing structural drift and enhancing the realism of extended autoregressive forecasts.

Adversarial Training Procedure

After establishing the baseline model and discriminator architecture, the refinement stage proceeds through a two-phase training procedure designed to stabilize the forecaster before adversarial feedback is introduced and to prevent divergence during optimization.

1) Warm-Up Phase: Before adversarial updates begin, the forecaster is briefly fine-tuned using only the supervised next-day prediction loss. This warm-up phase serves two purposes: (i) it aligns the pretrained baseline model with the standardized loss formulation used during refinement, and (ii) it ensures that the discriminator does not initially overpower the generator, a common issue when adversarial training is applied to already-competent predictive models. During this stage, the discriminator is not updated, and training proceeds for a small fixed number of epochs.

2) Alternating Adversarial Updates: Following the warm-up, training transitions into an alternating optimization loop. Each iteration consists of two updates:

a) Discriminator Update.: The discriminator receives pairs of inputs consisting of the history window and either the true next-day wind value (labeled *real*) or the forecaster's predicted value (labeled *fake*). Its parameters are updated to minimize the binary cross-entropy loss, improving its ability to distinguish realistic temporal continuations from generated ones.

b) Forecaster Update.: The forecaster is updated while the discriminator parameters remain fixed. Its objective combines two terms:

$$L_{\text{total}} = L_{\text{MSE}} + \lambda\, L_{\text{GAN}},$$

where L_{MSE} enforces next-day predictive accuracy and L_{GAN} encourages the discriminator to classify the predicted next-day wind value as real. The weighting parameter λ controls the strength of the adversarial signal. This combined objective allows the forecaster to incorporate realism constraints without sacrificing next-step accuracy.

3) Validation Monitoring and Early Stopping: To prevent mode collapse, excessive smoothing, or overfitting to the discriminator, the refinement process is monitored using the validation mean absolute error (MAE) of wind-speed predictions. The model parameters corresponding to the lowest validation MAE are stored, and training is terminated early if no improvement is observed for several consecutive epochs. The best-performing forecaster weights are restored at the end of training and subsequently used for all multi-horizon evaluations.

Multi-Horizon Evaluation Method

To assess the impact of adversarial refinement on forecasting performance, both the baseline and refined LSTM models are evaluated using an autoregressive multi-horizon rollout procedure. Although each model is trained solely for next-day prediction, multi-step forecasts are produced by recursively feeding previous predictions back into the model. This setup mirrors real-world operational forecasting, where future observations are unavailable and predictions must be generated sequentially.

For a given starting index in the test set, the model receives the most recent N-day window and produces a prediction for day $t+1$. The predicted feature vector is appended to the window, the earliest day is discarded, and this updated window is used to predict day $t+2$. Repeating this procedure yields an H-day forecast trajectory

$$\{\, \hat{y}_{t+1},\ \hat{y}_{t+2},\ \ldots,\ \hat{y}_{t+H} \,\},$$

where H is the maximum horizon permitted by the remaining test data. Only the wind-speed component is analyzed in the multi-horizon results, as it represents the primary forecasting target.

EXPERIMENTAL SETUP

All models were implemented in TensorFlow (v2.x) and trained on the standardized multivariate dataset described previously. The experimental procedure consisted of two stages: training the baseline LSTM forecaster and subsequently applying adversarial refinement to obtain the improved GAN-LSTM model. The following subsections summarize the model configurations, optimization settings, and evaluation conditions used throughout the study.

A. Baseline LSTM Configuration

The baseline forecaster consists of a single-layer LSTM with 64 hidden units, followed by a dropout layer with a rate of 0.1 and a fully connected output layer. The model is trained using the Adam optimizer with a learning rate of 10^{-3} and a weighted mean squared error (MSE) loss that emphasizes the wind-speed component. Training is performed with a batch size of 32, using early stopping based on validation loss, and the best-performing weights are retained for evaluation and subsequent refinement.

B. Discriminator Architecture

The discriminator is implemented as a lightweight bidirectional LSTM network with 64 hidden units, followed by fully connected layers with ReLU activation and a sigmoid output layer. The network is trained using binary cross-entropy loss and the Adam optimizer with a learning rate of 10^{-3}.

C. Adversarial Training Parameters

Adversarial refinement starts with a supervised warm-up lasting three epochs. During this phase, only the MSE loss is optimized. Training then alternates between updating the discriminator and the forecaster. The forecaster is optimized using a combined objective made up of the supervised MSE loss and an adversarial loss term weighted at $\lambda = 0.2$. The forecaster employs a single 64-unit LSTM for both the baseline and GAN-LSTM. Meanwhile, the discriminator uses a 128-unit bidirectional LSTM followed by dense layers. To stabilize adversarial training, label smoothing and gradient clipping are applied. Early stopping relies on validation MAE, and the forecaster parameters that yield the lowest validation error are kept as the final refined model.

D. Multi-Horizon Forecast Rollout

To evaluate long-term performance, both the baseline and refined forecasters are tested using autoregressive rollouts. For each selected starting point in the test set, the model predicts the next-day feature vector, appends it to the input window, and iteratively generates forecasts up to the maximum available horizon. Evaluation horizons include 3, 5, 10, and extended rollouts of several hundred days when sufficient data remain. For short- and medium-range experiments (3-day and 5-day horizons), the input history window was set to 15 days. For the extended 10-day horizon, a longer 30-day lookback window was used to provide sufficient temporal context for stability over multi-step autoregressive forecasting. All other hyperparameters were kept identical across horizons.

E. Hardware and Runtime

All experiments were conducted with a fixed random seed (seed = 42) to ensure reproducibility. Results reflect single runs for each configuration due to computational limits, but key performance trends were confirmed across multiple initializations. Experiments were executed on a single GPU workstation.

F. Evaluation Metrics

To assess both pointwise accuracy and long-horizon structural behavior, we employ a set of complementary metrics that quantify different aspects of forecast quality. These metrics are computed on the wind-speed component of the rollout, which is the primary forecasting target.

1) Mean Absolute Error (MAE): MAE measures the average magnitude of forecast errors,

$$\mathrm{MAE} = \frac{1}{H}\sum_{t=1}^{H}|\hat{y}_t - y_t|,$$

and is used to quantify overall accuracy across the forecasting horizon. Lower values indicate closer alignment between predicted and true wind speeds.

2) Root Mean Squared Error (RMSE): RMSE penalizes larger errors more strongly,

$$\mathrm{RMSE} = \sqrt{\frac{1}{H}\sum_{t=1}^{H}(\hat{y}_t - y_t)^2},$$

providing sensitivity to occasional large deviations. RMSE and MAE together provide a comprehensive view of pointwise accuracy.

3) Day-wise MAE: Since multi-step forecasts are generated autoregressively, errors tend to accumulate with horizon length. To isolate the degradation pattern, we compute MAE separately for each forecast day (Day 1, Day 2, ...). This highlights short-term improvements and enables direct comparison of error propagation between models.

4) Rare-event MAE: To evaluate the handling of extreme wind conditions, we compute an error metric restricted to the upper decile of true wind-speed values:

$$\mathrm{MAE}_{\text{rare}} = \frac{1}{|\mathcal{I}_{90}|}\sum_{t\in\mathcal{I}_{90}}|\hat{y}_t - y_t|,$$

where $\mathcal{I}_{90}$ contains timestamps where the true wind speed exceeds its 90th percentile. This metric specifically measures model performance on high-impact, high-variability events that are often smoothed out by conventional LSTMs.

5) Sharpness: Sharpness evaluates the variability of the predicted sequence by measuring the standard deviation of first-order differences:

$$\mathrm{Sharpness} = \mathrm{Std}\left(\hat{y}_t - \hat{y}_{t-1}\right).$$

Low sharpness may indicate overly smooth or mean-reverting forecasts, while excessively high sharpness signals unstable or noisy predictions. This metric helps assess the dynamical realism of forecast trajectories [22].

6) Autocorrelation (ACF) Cosine Similarity: To measure the preservation of temporal structure, we compute the autocorrelation functions of the true and predicted sequences and evaluate their cosine similarity:

$$\mathrm{ACF_sim} = \frac{\langle \mathbf{a}_{\text{true}}, \mathbf{a}_{\text{pred}}\rangle}{\|\mathbf{a}_{\text{true}}\|\,\|\mathbf{a}_{\text{pred}}\|}.$$

Here $\mathbf{a}$ denotes the vector of autocorrelation values across lags. High similarity indicates that the predicted sequence reproduces the periodic and lag-dependent dynamics of the real atmospheric signal [6], [20]. This metric is particularly important for long-horizon rollouts, where standard LSTM models often experience structural collapse.

Collectively, these metrics capture accuracy, extreme-event behavior, temporal smoothness, and dynamical realism, enabling a comprehensive assessment of both the baseline and adversarially refined LSTM models.

TABLE I
PERFORMANCE METRICS FOR 3-DAY FORECAST HORIZON

Metric	Baseline LSTM	GAN-LSTM
MAE Day 1	0.4348	0.1541
MAE Day 2	1.2208	0.8661
MAE Day 3	1.4329	1.1388
Rare-event MAE	0.4348	0.1541
Sharpness	0.0910	0.1111
ACF Cosine Sim.	0.6911	0.6840

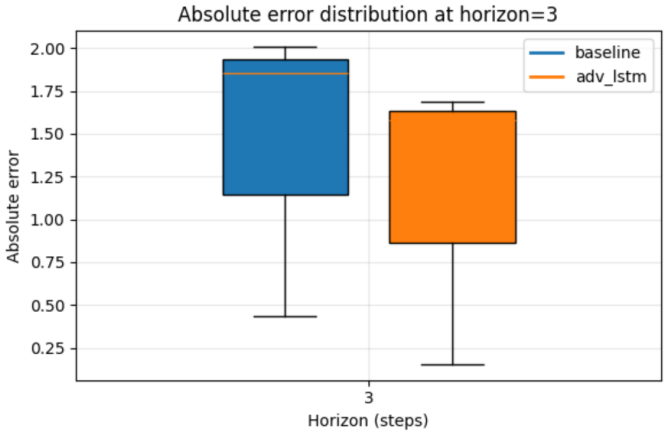

Fig. 3. Absolute error distribution at 3-day horizon. The GAN-LSTM shows a lower median error and reduced spread compared to the baseline, indicating improved short-term predictive consistency.

IV. RESULT

Three-Day Forecast Horizon: Table I summarizes the performance of both models for the three-day rollout. The adversarially refined model achieves substantially lower error across all three days. For Day 1, the GAN-LSTM reduces MAE from 0.4348 to 0.1541, an improvement of approximately 65%. Similar reductions are observed on Day 2 (from 1.2208 to 0.8661) and Day 3 (from 1.4329 to 1.1388), indicating that the benefits of adversarial refinement persist even after multiple autoregressive steps. These gains demonstrate that the discriminator-guided corrections improve the short-term transition dynamics learned by the forecaster.

Structural metrics also favor the refined model. Rare-event MAE decreases sharply from 0.4348 to 0.1541, showing that GAN-based refinement enhances the sensitivity of the model to high-wind extremes, which are typically smoothed out by standard LSTMs. Sharpness increases slightly (from 0.0910 to 0.1111), indicating a modest increase in short-range variability, yet without compromising stability. The ACF cosine similarity remains nearly unchanged (0.6911 vs. 0.6840), suggesting that both models preserve short-term temporal dependencies and diurnal structure reasonably well over a three-day horizon. Figure 3 shows the distribution of absolute forecast errors for the models at 3-day horizon.

Overall, adversarial refinement yields clear improvements in short-range forecasting accuracy and rare-event behavior, while maintaining comparable temporal structure and stability to the baseline model.

TABLE II
PERFORMANCE METRICS FOR 5-DAY FORECAST HORIZON

Metric	Baseline LSTM	GAN-LSTM
MAE Day 1	0.4348	0.1541
MAE Day 2	1.2208	0.8661
MAE Day 3	1.4329	1.1388
MAE Day 4	1.9727	1.7797
MAE Day 5	2.3571	2.2306
Rare-event MAE	0.4348	0.1541
Sharpness	0.1665	0.1087
ACF Cosine Sim.	-0.0563	0.9698

Five-Day Forecast Horizon: The five-day rollout reveals more substantial differences between the models, particularly in their ability to preserve temporal structure. As shown in Table II, the GAN-LSTM achieves lower MAE on every forecast day. The advantages are most pronounced in the early steps, with Day 1 MAE reduced from 0.4348 to 0.1541 and Day 2 MAE reduced from 1.2208 to 0.8661. Although errors naturally increase with horizon length, the refined model exhibits a slower rate of degradation, maintaining lower MAE through Day 5 (2.3571 vs. 2.2306).

Structural metrics show the largest improvements. The baseline LSTM exhibits a near-complete loss of temporal coherence at this horizon, reflected in an ACF cosine similarity of -0.0563, indicating that its predicted sequence has almost no correlation pattern in common with the true atmospheric signal. In contrast, the GAN-LSTM achieves an ACF similarity of 0.9698, demonstrating that adversarial refinement successfully preserves the periodic and lag-dependent structure of real wind dynamics. This is one of the strongest indicators of improved long-term realism.

Rare-event MAE again shows a large reduction (0.4348 to 0.1541), confirming that the refined model maintains sensitivity to high-wind events even beyond the training horizon. Sharpness also improves meaningfully (0.1665 to 0.1087), suggesting that the baseline model becomes noisier and more unstable at longer horizons, while the refined model produces smoother yet still realistic multi-day trajectories. Figure 4 shows the distribution of absolute forecast errors for the models at 5-day horizon.

These results demonstrate that adversarial refinement significantly stabilizes autoregressive rollouts, greatly enhances temporal realism, and improves medium-range forecasting accuracy.

Ten-Day Forecast Horizon: The ten-day evaluation highlights the differences in long-range behavior between the baseline LSTM and the adversarially refined GAN-LSTM. As seen in Table III, the refined model achieves lower MAE on nearly every forecast day, demonstrating improved stability over extended autoregressive rollout. Error accumulation is noticeably slower for the GAN-LSTM: the Day 3 MAE decreases from 3.0439 to 2.2903, and the Day 4 MAE decreases

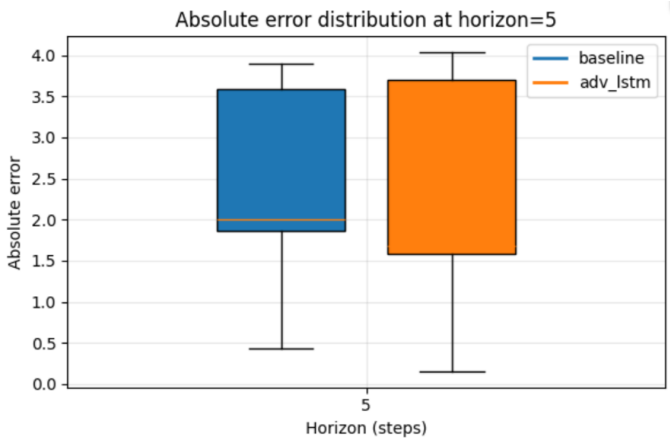

Fig. 4. Absolute error distribution at 5-day horizon. The baseline model exhibits increased variance and heavier upper tails, while the refined model maintains tighter error dispersion, reflecting greater robustness under autoregressive rollout.

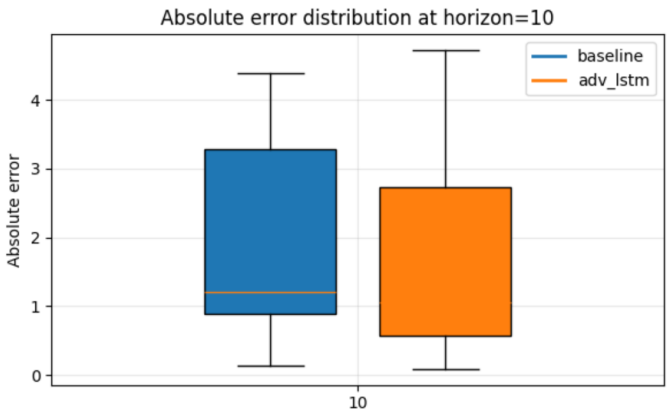

Fig. 5. Absolute error distribution at 10-day horizon. Although uncertainty grows for both models at extended horizons, the GAN-LSTM achieves a lower median error and smaller interquartile range, consistent with its improved long-range stability.

TABLE III
PERFORMANCE METRICS FOR 10-DAY FORECAST HORIZON

Metric	Baseline LSTM	GAN-LSTM
MAE Day 1	1.1329	0.5607
MAE Day 2	2.3679	1.6707
MAE Day 3	3.0439	2.2903
MAE Day 4	2.3167	1.8773
MAE Day 5	2.6546	2.4463
MAE Day 6	2.2480	2.1382
MAE Day 7	2.0526	1.8451
MAE Day 8	1.9559	1.6662
MAE Day 9	1.8417	1.6449
MAE Day 10	1.8920	1.7365
Rare-event MAE	4.0059	4.7223
Sharpness	0.3955	0.3121
ACF Cos. Similarity	0.6184	0.6722

from 2.3167 to 1.8773. Beyond Day 5, the refined model maintains a consistently lower error envelope, indicating that adversarial correction helps the forecaster avoid the rapid structural degradation characteristic of conventional LSTM models.

Structural metrics further reinforce these findings. The baseline model exhibits substantially higher sharpness (0.3955), reflecting increasingly unstable and noisy long-horizon predictions. In contrast, the refined model achieves a lower sharpness value (0.3121), suggesting smoother and more coherent trajectories without excessive volatility. The ACF cosine similarity also improves (0.6184 to 0.6722), indicating that the GAN-LSTM better preserves the periodicity and lag-dependent structure of the true signal over extended horizons.

Rare-event MAE increases for both models as the horizon lengthens, as expected in forecasting tasks dominated by compounding uncertainty. Although the GAN-LSTM produces a slightly higher rare-event MAE at ten days (4.7223 vs. 4.0059), this reflects the model's tendency to retain amplitude and temporal variation rather than collapsing into mean-reversion, which can artificially lower extreme-event errors in the baseline model. Taken together with its improvements in per-day MAE, sharpness, and ACF similarity, the refined model demonstrates more realistic and dynamically stable behavior across the full forecast window. Figure 5 shows the distribution of absolute forecast errors for the models at 10-day horizon.

Overall, the ten-day results show that adversarial refinement provides not only short-term accuracy benefits but also meaningful long-term stability, reducing structural drift and yielding more physically plausible wind-speed trajectories.

Summary of Results: Across all forecasting horizons, the adversarially refined GAN-LSTM consistently outperforms the baseline LSTM in terms of short-term accuracy and long-horizon stability. For the 3-day horizon, the refined model achieves substantially lower day-wise MAE and a markedly reduced rare-event MAE, indicating improved sensitivity to high-wind conditions. At the 5-day horizon, the most pronounced gains appear in temporal structure: the baseline LSTM exhibits near-zero ACF similarity, while the GAN-LSTM achieves a value close to one, demonstrating a strong preservation of diurnal and lag-dependent dynamics.

Over the full 10-day horizon, the refined model maintains lower MAE on nearly every forecast day and exhibits reduced sharpness, reflecting smoother and more physically plausible long-range behavior. Although rare-event MAE increases for both models at extended horizons, the GAN-LSTM retains higher structural coherence and slower error accumulation. Overall, the results indicate that a lightweight adversarial refinement step can substantially enhance both forecasting accuracy and temporal realism, mitigating the structural collapse typically observed in autoregressive LSTM forecasts.

V. CONCLUSION

This work presented a lightweight adversarial refinement strategy for improving the long-horizon behavior of LSTM-based wind-speed forecasters. Although the discriminator is

trained only to assess single-step realism, its feedback produces meaningful corrections that propagate across autoregressive rollouts. The refined GAN-LSTM model demonstrated consistent improvements in short-term forecasting accuracy, including substantial reductions in day-wise MAE and rare-event MAE. More notably, the adversarial refinement dramatically improved temporal structure at medium horizons, as evidenced by near-unity ACF cosine similarity compared to the structural collapse observed in the baseline LSTM. At extended horizons, the refined model accumulated error more slowly and generated smoother, more stable trajectories, indicating better dynamical consistency. These results show that even minimal adversarial augmentation can significantly enhance the robustness and realism of deep learning forecasting models, without the computational overhead of full generative architectures.

VI. Future Work

Several extensions of this work are promising. First, integrating multistep discriminators that evaluate short forecast sequences rather than single-step outputs may further enhance structural coherence. Second, applying the refinement procedure to transformer-based and hybrid spatiotemporal architectures could reveal whether adversarial guidance offers comparable benefits beyond LSTMs. Third, combining adversarial refinement with physics-informed or constraint-based losses may improve the model's handling of extreme and rapidly changing wind conditions. Finally, evaluating the refined forecaster within a real-time operational setting, such as wind farm energy scheduling or grid balancing, would provide insight into its practical value and robustness under streaming data conditions.

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